Attention-based Pairwise-Semantic Fusion for YouTube Clickbait Detection

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Abstract

As the popularity of YouTube continues to grow, competition for attracting views has intensified, leading to a rapid increase in clickbait content that induces user clicks through misleading titles and thumbnails. As a result, users frequently encounter videos whose actual content does not align with the expectations formed by the title and thumbnail. To address this issue, this study proposes a multimodal clickbait detection framework that integrates linguistic information (video titles and automatic speech recognition (ASR) transcripts) and visual information (thumbnails and keyframes). Visual information is first converted into textual representations through image captioning, and all textual data are embedded using Sentence-BERT (SBERT). Subsequently, attention pooling is applied to the transcript and keyframe sequences using the title and thumbnail as queries, and modality pairs are constructed to capture the semantic inconsistency between expected information and actual content. Experimental results demonstrate that the proposed approach improves clickbait detection by effectively capturing semantic relationships between titles, thumbnails, and actual video content.

Keywords: Clickbait Detection, Semantic Inconsistency, Multimodal, Attention Pooling

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1. Introduction

Recently, online video platforms such as YouTube have become one of the most influential media for information consumption and communication worldwide. As these platforms have rapidly grown, competition for views and advertising revenue has intensified, leading to the widespread adoption of various forms of clickbait. Clickbait refers to strategies that induce clicks through sensational or exaggerated titles and thumbnails. While such practices may temporarily increase view counts, they ultimately undermine content credibility, induce user fatigue, and may result in user disengagement from the platform. Against this backdrop, understanding and detecting the semantic inconsistency between titles or thumbnails and the actual video content has emerged as an important research problem. The negative perception users experience when encountering clickbait does not primarily stem from exaggeration or emotional expression itself. Instead, the core issue lies in the semantic inconsistency between the expectations formed by the title or thumbnail and the content delivered by the video.

However, existing studies have largely defined clickbait in terms of stylistic linguistic features such as exaggerated expressions, emotional language, sensational wording, or references to celebrities, and have focused on detecting these characteristics [1,2]. Since such stylistic elements are also commonly used in general content to attract viewer interest, relying

solely on the presence of specific expressions is insufficient for reliably determining content distortion. This expression-centered perspective has introduced methodological bias in clickbait detection. Prior research has tended to rely excessively on linguistic signals such as titles or ASR transcripts [3], while relatively neglecting visual information such as thumbnails or video frames. In addition, several studies have attempted to utilize statistical metadata such as view counts or like–dislike ratios [4,5]; however, such indicators are only available after content consumption, which makes them unsuitable for preventive detection at the time of upload. Although recent studies have explored multimodal approaches that jointly consider text and images, they still exhibit limitations in capturing meaningful semantic relationships across modalities. Existing methods typically encode text and images independently and then fuse them using element-wise averaging [5]. This strategy fails to precisely model the subtle semantic conflicts between the "expected information" conveyed by titles or thumbnails and the "actual information" presented in the video content. Consequently, the semantic gap between expectations and reality is not effectively captured.

To address these limitations, we propose a multimodal framework that leverages four modalities: title, thumbnail, transcript, and keyframe. By combining these modalities, we construct four semantic comparison pairs: title–transcript, title–keyframe, thumbnail–transcript, and thumbnail–keyframe. Each pair is designed to directly extract semantic relationships between modalities. The resulting representations are integrated through a branch-level attention mechanism, enabling the model to learn the relative importance of each modality pair. This process produces a unified representation that reflects which modality interactions contribute most to clickbait detection, and this representation is used for final prediction.

Experimental results show that our framework effectively captures semantic relationships between titles, thumbnails, and actual video content.

The main contributions of this work are summarized as follows:

- First, we propose addressing clickbait detection from the perspective of **semantic inconsistency**, emphasizing the relationship between expectations formed by titles and thumbnails and the actual video content.
- Second, we present a multimodal architecture that systematically constructs **four semantic comparison pairs** using title, thumbnail, transcript, and keyframe modalities.
- Finally, we introduce a **branch-level attention-based integration mechanism** that captures fine-grained semantic interactions across modality pairs and effectively aggregates critical information for clickbait prediction.

2. Data Construction

This study constructed a multimodal dataset based on news videos to quantitatively analyze the semantic relationships among titles, thumbnails, transcripts, and keyframes.

2.1 Data Collection and Artificial Clickbait Generation

We collected short news videos of approximately three minutes in length from the official YouTube channels of major Korean broadcasters, including SBS[1], KBS[2], MBC[3], and YTN[4]. The collected videos were limited to content such as celebrity news, social issues, and interviews, which contain rich visual and linguistic cues suitable for evaluating semantic inconsistency-based clickbait detection. Samples with poor audio quality, low resolution, or duplicated titles or thumbnails were filtered out in advance. Collecting enough real-world clickbait videos exhibiting semantic inconsistency is challenging because such videos are relatively rare, may be quickly removed due to platform policies, and require human reviewers to inspect the full video content. To overcome these limitations, this study adopts a strategy of artificially generating clickbait samples. Specifically, non-clickbait videos with similar topics are paired, and intentional semantic inconsistencies are introduced by swapping titles and thumbnails within each pair. The pairing was manually performed based on human judgment of topical similarity; for example, videos (0, 1), (2, 3), and (4, 5) were grouped as topically similar pairs. For each video v_i, the title, thumbnail, transcript, and keyframes were extracted.

The transcript generated via automatic speech recognition is hereafter referred to as STT (Speech-to-Text). Prior to embedding, the original non-clickbait videos constituting each pair (v_{2i}, v_{2i+1}) are defined as follows: $v_{2i} = (title_{2i}, thumbnail_{2i}, STT_{2i}, keyframe_{2i})$ and $v_{2i+1} = (title_{2i+1}, thumbnail_{2i+1}, STT_{2i+1}, keyframe_{2i+1})$.

Artificial clickbait samples were generated as $v_{2i} = (title_{2i+1}, thumbnail_{2i+1}, STT_{2i}, keyframe_{2i})$ or $v_{2i+1} = (title_{2i}, thumbnail_{2i}, STT_{2i+1}, keyframe_{2i+1})$ In this process, the actual video content, namely the STT and keyframes, remains unchanged, while only the expectation-

[1] https://www.youtube.com/@sbsnews8
[2] https://www.youtube.com/@newskbs
[3] https://www.youtube.com/@MBCNEWS11
[4] https://www.youtube.com/@ytnstar

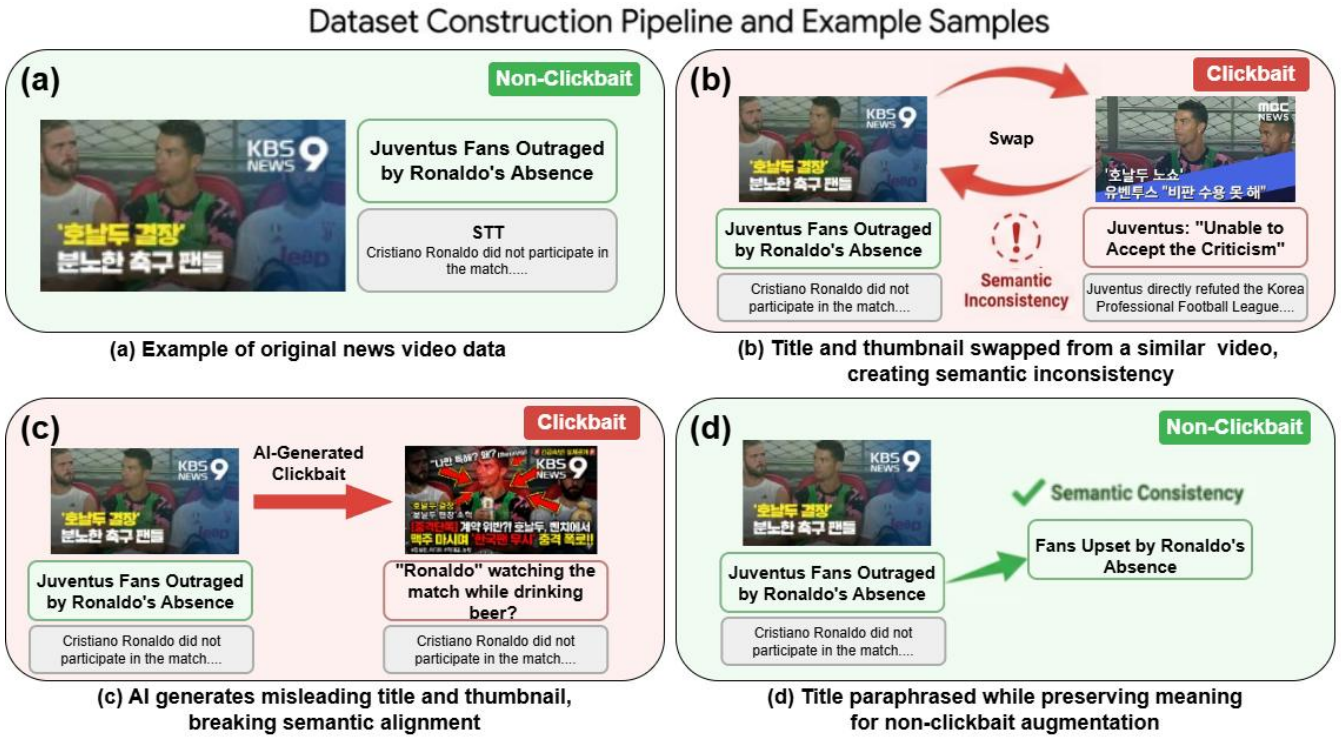


Figure 1. Example of Dataset Construction Methods

setting elements (titles and thumbnails) are replaced. This procedure introduces intentional semantic inconsistency between the expected information and the actual video content without modifying the underlying video itself. In addition to the swap-based construction, we employed an AI-based generation strategy to introduce more diverse forms of semantic inconsistency. GPT-5 was provided with the original video title and STT and prompted to generate a clickbait-style title that was semantically inconsistent with the actual video content. Based on the generated title, Gemini Nano Banana 2 was then used to generate a corresponding clickbait thumbnail aligned with the misleading title. This process complements simple modality swapping by introducing semantic inconsistencies generated through generative models.

For non-clickbait samples, we aimed to prevent the model from relying on superficial correlations between specific lexical patterns and clickbait labels. To this end, the original titles were paraphrased while preserving their semantic meaning. This approach maintains semantic consistency while increasing linguistic diversity, thereby encouraging the model to focus on semantic alignment between the expectation-setting information and the actual video content rather than on surface-level lexical patterns. Figure 1 illustrates the overall dataset construction process, including both swap-based augmentation and AI-based clickbait generation. The detailed dataset composition and experimental splits are provided in Section 4.1.

2.2 Modality-specific Feature Extraction and Textual Unification

To enable comparison and learning across all modalities within a shared semantic space, we unified all modality representations into a text-based format.

a) Thumbnail captioning

For each video, the thumbnail image was captioned using the Qwen2-VL-7B-Instruct model [6] together with the corresponding title, converting visual information into a textual representation.

b) Title preprocessing

Video titles were refined by removing channel-specific tokens (e.g., broadcaster names) and unnecessary special characters to normalize the textual representation.

c) STT generation

The audio signal of each video was converted into text using the Whisper ASR model [7]. Through these preprocessing steps, thumbnails, titles, and STT were unified as text-based inputs, allowing them to be compared and integrated within a shared semantic space during the embedding stage.

d) Keyframe extraction and scene captioning

To structurally represent video content, a four-stage procedure was applied to extract keyframes at the scene level and convert

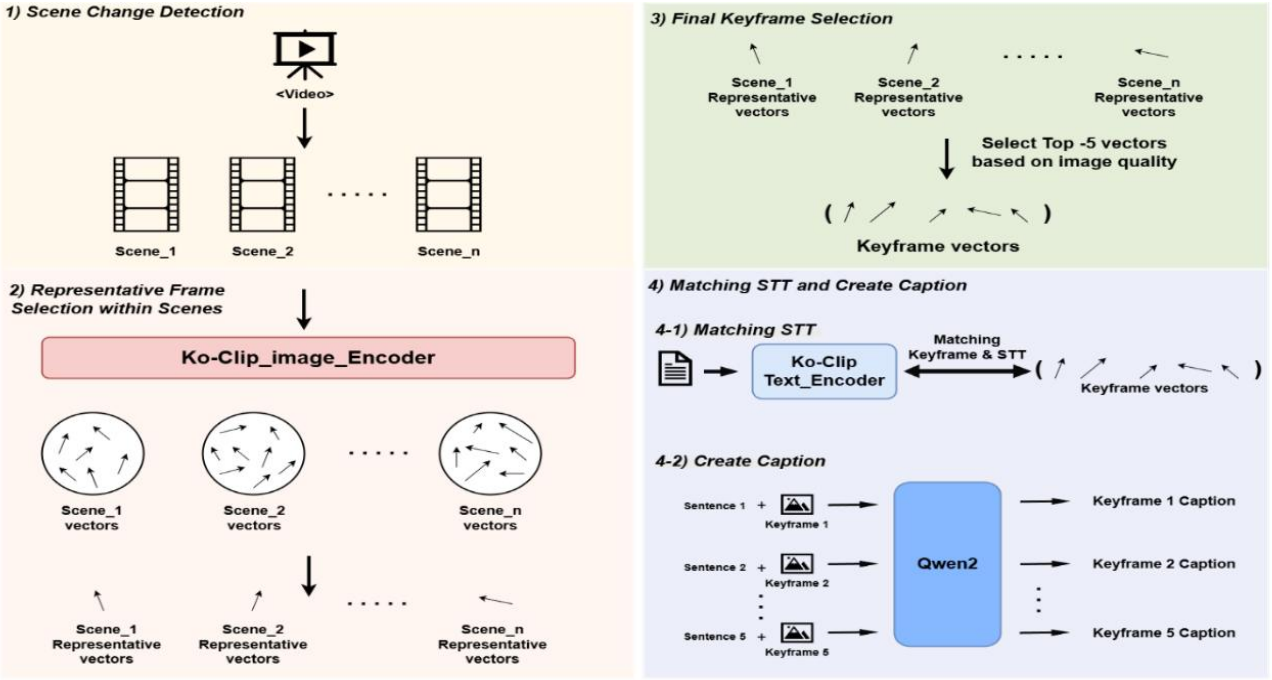


Figure 2. Overall pipeline for generating scene-level captions from video

them into textual descriptions.

Scene change detection: Scene boundaries were defined based on abrupt differences in color histograms and pixel variations between consecutive frames. Since scene boundaries correspond to points of visual change in the video, selecting keyframes based on these boundaries reduces redundancy while preserving representative visual information, making this approach well suited for semantic inconsistency analysis.

Representative frame selection within scenes: Within each scene segment, frames were embedded using the KoCLIP image encoder [8]. The frame whose embedding was closest to the mean embedding of the segment was selected as the representative frame for that scene.

Final keyframe selection: Among the representative frame candidates, the top five frames with the highest image quality scores were selected as the final keyframes.

Matching STT and caption generation: Cosine similarity was computed between each extracted keyframe image and the STT sentences, and the most semantically relevant STT sentence was matched to each keyframe. These matched pairs were then provided as input to the Qwen2-VL-7B-Instruct model to generate visual cues with linguistic context.

Figure 2 illustrates the overall keyframe captioning pipeline, including scene change detection, keyframe selection, STT matching, and caption generation. Through this process, titles, audio, thumbnails, and keyframes were all converted into textual representations. Finally, all textual inputs were embedded using Sentence-BERT.

3. Multimodal Framework for Semantic Consistency

3.1 Overview

We propose a novel multimodal model that directly compares the semantic consistency between the expectations presented by video titles and thumbnail captions and the actual video content, represented by STT and keyframe captions. The proposed model embeds four input modalities—title, thumbnail caption, STT and keyframe captions—into a shared semantic space and then employs an attention-based mechanism to extract semantically important segments from the video content (STT and keyframe captions) with respect to the title and thumbnail caption.

Subsequently, relational representations are constructed for each of the four modality pairs, and these signals are integrated through branch-level attention to ultimately determine whether the video exhibits clickbait characteristics. Figure 3 illustrates the overall architecture of the proposed framework.

3.2 Attention Pooling and Pairwise Relation Modeling

The proposed model utilizes four input modalities—title, thumbnail caption, STT and keyframe captions. Each input is transformed into a 768-dimensional embedding vector ($\boldsymbol{h} \in \mathbb{R}^{768}$)

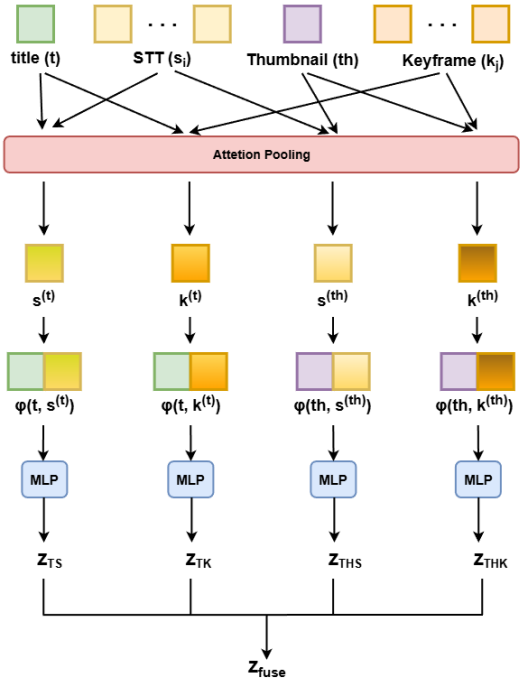


Figure 3. Overview of the proposed Pairwise-Semantic Fusion architecture

using a Sentence-BERT (SBERT) model, and L2 normalization is applied to ensure stability in cosine similarity computation. The title and thumbnail caption are represented as single vectors ($\boldsymbol{t}$, $\boldsymbol{th}$), whereas the STT and keyframe captions are represented as variable-length sequences ($\{\boldsymbol{s}_i\}$, $\{\boldsymbol{k}_j\}$).

Attention Pooling

Since the STT and keyframe captions consist of multiple tokens, an attention-pooling mechanism is employed to aggregate the element embeddings into a single vector [9]. In this process, the title ($\boldsymbol{t}$) or thumbnail ($\boldsymbol{th}$) embedding is used as the query. Here $\boldsymbol{x_i}$ denotes a generic element embedding from the input sequence, which corresponds to either an STT sentence ($\boldsymbol{s}_i$) or a keyframe caption ($\boldsymbol{k}_j$), depending on the modality. For each sequence element $\boldsymbol{x}_i$ and query $\boldsymbol{q} \in \{\boldsymbol{t}, \boldsymbol{th}\}$, a cosine similarity score $\boldsymbol{u}_i$ is computed, and the corresponding attention weight $\boldsymbol{\alpha}_i$ is obtained by applying SoftMax normalization, as defined in the following equation [9].

$$\boldsymbol{u_i} = \frac{\cos(\boldsymbol{x_i}, \boldsymbol{q})}{\boldsymbol{\tau}}, \boldsymbol{\alpha_i} = \frac{\exp(\boldsymbol{u_i})}{\sum_j \exp(\boldsymbol{u_j})} \quad (1)$$

where $\boldsymbol{\tau}$ is a temperature parameter controlling the sharpness of the distribution. The aggregated vector is then obtained as

$$\bar{\boldsymbol{x}} = \sum_i \boldsymbol{\alpha_i x_i} \quad (2)$$

Through this process, the model uses the title or thumbnail caption as a query and captures the semantic relevance to the corresponding STT sentences and keyframe caption sequences of the video content, thereby transforming variable-length sequences into fixed-length semantic vectors.

Pairwise Feature Construction

To explicitly capture relational semantics across modalities, two embeddings $\boldsymbol{h_1}$ and $\boldsymbol{h_2}$ are combined to form a pairwise feature vector [10]:

$$\boldsymbol{\varphi(h_1, h_2)} = [\, \boldsymbol{h_1}; \boldsymbol{h_2}; |\, \boldsymbol{h_1} - \boldsymbol{h_2}\, |; \boldsymbol{h_1} \odot \boldsymbol{h_2}\,] \in \mathbb{R}^{\boldsymbol{4d}} \quad (3)$$

This formulation encodes both the semantic difference between the two embeddings ($|\, \boldsymbol{h_1} - \boldsymbol{h_2}\, |$) and their element-wise interactions ($\boldsymbol{h_1} \odot \boldsymbol{h_2}$). In total, four pairwise branch representations are constructed: $\boldsymbol{\varphi(t, s^{(t)})}$ for title–STT, $\boldsymbol{\varphi(t, k^{(t)})}$ for title–keyframe captions, $\boldsymbol{\varphi(th, s^{(th)})}$ for thumbnail caption–STT, and $\boldsymbol{\varphi(th, k^{(th)})}$ for the thumbnail caption–keyframe captions relationships. All branch inputs are normalized using Layer Normalization to ensure numerical stability.

3.3 Branch-Level Attention Fusion

The outputs of the four previously generated branches are combined into a single matrix ***Z:***

$$\boldsymbol{Z} = [\boldsymbol{z_{TS}}, \boldsymbol{z_{TK}}, \boldsymbol{z_{ThS}}, \boldsymbol{z_{ThK}}] \in \mathbb{R}^{\boldsymbol{4\times 4d}} \quad (4)$$

A learnable attention weight matrix $\boldsymbol{W_a} \in \mathbb{R}^{\boldsymbol{4d\times 1}}$ is applied to compute the contribution of each branch, and the resulting scores are normalized using a SoftMax function to obtain attention weights $\boldsymbol{\alpha_b}$.

$$\boldsymbol{\alpha_b} = \frac{\exp(\boldsymbol{W_a^\top z_b})}{\sum_{\boldsymbol{b'}} \exp(\boldsymbol{W_a^\top z_{b'}})} \quad (5)$$

The final fused representation is defined as the weighted sum of the branch features:

$$\tilde{\boldsymbol{z}} = \sum_{\boldsymbol{b}} \boldsymbol{\alpha_b z_b} \quad (6)$$

This vector is then passed through a linear projection layer to produce the final output logit $\boldsymbol{s_{fuse}}$. Through this process, the

model adaptively assigns higher weights to modality pairs that are more informative for clickbait detection. The total loss is computed by combining branch-level logits and the fused logit using weighted binary cross-entropy (BCE) loss.

$$\mathcal{L} = \lambda_{br} \sum_{b} \mathrm{BCE}(s_b, \hat{y}) + \mu\, \mathrm{BCE}(s_{fuse}, \hat{y}) \quad (7)$$

Here, $\hat{y}$ denotes the target label with smoothing applied. During inference, the sigmoid function is applied to the fused logit, and $\sigma(s_{fuse})$ is used as the final prediction probability. The resulting branch-level attention weights α_b provide a quantitative measure of how much each modality pair (e.g., title–STT, thumbnail caption–keyframe captions) contributes to the clickbait prediction.

4. Experiment

4.1 Setup

Data

For training and evaluating the proposed model, we constructed a dataset consisting of 800 multimodal samples following the procedure described in Section 2. The dataset was balanced, comprising 400 clickbait and 400 non-clickbait samples. Among the clickbait samples, 200 were generated by swapping the titles and thumbnails between videos with similar topics, while the remaining 200 were generated using the AI-based method. The non-clickbait samples consisted of 200 original news videos and 200 augmented samples generated by paraphrasing the original titles while preserving their semantic meaning. The 800 samples were divided into training, validation, and test sets containing 560, 120, and 120 samples, respectively. Each sample consists of four modalities: title, thumbnail caption, STT, and keyframe captions, each of which was embedded using SBERT. In the proposed model, four semantic comparison pairs were constructed: title–STT (TS), title–keyframe (TK), thumbnail–STT (ThS), and thumbnail–keyframe (ThK). These pairs were used to model the semantic relationships between the information conveyed by the title and thumbnail and the actual video content.

In addition to the constructed dataset, we employed *YTClickbait21K* [11], a publicly available dataset consisting of real YouTube videos collected across diverse content domains, to evaluate the generalization ability of the proposed model in real-world environments. Considering the focus of this study on short video length, we selected videos approximately five minutes or shorter. Among these videos, 150 clickbait samples exhibiting semantic inconsistencies between the title or thumbnail and the actual video content were selected, along with 150 non-clickbait samples, resulting in a balanced real-world set of 300 samples. The real-world dataset was divided into a validation set of 100 samples and a test set of 200 samples. The validation set was used only to determine the decision threshold, while the test set was used for the final performance evaluation. None of the real-world samples were used for model training. This setting was used to examine whether the model trained on the constructed dataset could generalize to semantic inconsistency-based clickbait occurring in real-world YouTube content across diverse content domains.

Implementation details

We trained the model using AdamW with a learning rate of $\mathbf{1 \times 10^{-3}}$ and a weight decay of $\mathbf{1 \times 10^{-3}}$. A ReduceLROnPlateau scheduler (factor = 0.5, patience = 3) was applied based on the validation balanced accuracy. Training was conducted for up to 80 epochs with a batch size of 32, and early stopping with a patience of 12 was employed. Each branch MLP has a hidden dimension of 64 with a dropout rate of 0.4. The SBERT encoder was kept frozen and was not fine-tuned during training. The training objective was defined as a weighted combination of the binary cross-entropy loss for the final fused prediction and the binary cross-entropy loss for the branch-level predictions, with weights of $\boldsymbol{\mu = 1.5}$ and $\boldsymbol{\lambda_{br} = 0.5}$, respectively. The decision threshold for classification was tuned on the validation set and subsequently applied to the corresponding test set. The same procedure was applied to the real-world evaluation. All experiments were independently repeated 20 times with different random seeds. The final results are reported as the mean and standard deviation (mean ± SD) across the 20 runs. Model performance was evaluated using Area Under the ROC Curve (AUC), Accuracy (ACC) and F1-score (F1).

4.2 Main Results

To evaluate the performance of the proposed method, we constructed five baselines employing different multimodal fusion and semantic relationship modeling strategies. **Independent-Pairwise Branch** constructs four pairwise branches—TS, TK, ThS, and ThK—and processes each branch independently using an MLP classifier. The final prediction is obtained by averaging the branch-level prediction scores without explicitly integrating information across branches. **Unimodal Score Averaging** [5] independently processes the title, thumbnail caption, STT, and keyframe captions using separate MLP classifiers and obtains the final prediction by averaging the four modality-level prediction scores. Unlike the pairwise approaches, this baseline does not

explicitly model cross-modal semantic relationships. **Cosine Similarity** uses the four pairwise cosine similarity scores (TS, TK, ThS, and ThK) as input features for clickbait classification, examining whether explicit similarity scores alone are sufficient to capture semantic relationships. **Cross-Attention** represents the title, thumbnail caption, STT, and keyframe captions as modality tokens and applies a Transformer-based attention encoder to model interactions among them. The resulting multimodal representation is then used for final clickbait classification. **Causal Representation (adapted)** [12] decomposes the multimodal representation into invariant and variant components and performs prediction primarily using the invariant representation, providing a robustness-oriented representation learning baseline. These baselines enable comparison with unimodal prediction, explicit pairwise modeling, similarity-based modeling, attention-based multimodal interaction, and robustness-oriented representation learning. For a fair comparison, all baselines and the proposed method used the same four modalities: title, thumbnail, STT, and keyframe. The same data splits and evaluation settings were also applied to all methods, allowing performance differences to be attributed to differences in modality representation and fusion strategies rather than to differences in the input information.

4.2.1 Results on Synthetic Data

Table 1. Comparison of fusion strategies in Synthetic Data

Method	AUC	ACC	F1
Independent-Pairwise Branch	0.775 ± 0.006	0.717 ± 0.014	0.703 ± 0.026
Unimodal Score Averaging	0.745 ± 0.019	0.671 ± 0.056	0.706 ± 0.022
Cosine Similarity	0.774 ± 0.016	0.696 ± 0.039	0.656 ± 0.086
Cross-Attention	0.748 ± 0.023	0.696 ± 0.033	0.685 ± 0.026
Causal Representation (adapted)	0.784 ± 0.009	0.723 ± 0.021	0.718 ± 0.027
Pairwise-Semantic Fusion (ours)	**0.857 ± 0.018**	**0.735 ± 0.038**	**0.766 ± 0.024**

Table 1 presents the performance comparison between the proposed Pairwise-Semantic Fusion and the five baseline methods described in Section 4.2. Pairwise-Semantic Fusion outperforms all five baselines across all three metrics, achieving an AUC of 0.857, an ACC of 0.735, and an F1-score of 0.766. Although the ACC margin over the strongest baseline, Causal Representation (adapted), is small (0.735 vs. 0.723), the AUC and F1-score gains are more pronounced (0.857 vs. 0.784 and 0.766 vs. 0.718). These results indicate that learning relational representations across modalities effectively captures semantic inconsistency, which is the core factor in clickbait detection.

4.2.2 Results on Real-World Data

Table 2. Comparison of fusion strategies in Real-World Data

Method	AUC	ACC	F1
Independent-Pairwise Branch	0.713 ± 0.041	0.634 ± 0.039	0.592 ± 0.092
Unimodal Score Averaging	0.713 ± 0.038	0.637 ± 0.035	0.647 ± 0.103
Cosine Similarity	0.824 ± 0.085	0.755 ± 0.079	0.753 ± 0.078
Cross-Attention	0.713 ± 0.054	0.640 ± 0.039	0.641 ± 0.057
Causal Representation (adapted)	0.672 ± 0.035	0.617 ± 0.032	0.602 ± 0.048
Pairwise-Semantic Fusion (ours)	**0.851 ± 0.037**	**0.757 ± 0.033**	**0.766 ± 0.048**

Table 2 presents the performance comparison between the proposed method and the baseline methods on real-world YouTube data. Pairwise-Semantic Fusion achieved the highest average performance across all three evaluation metrics, with an AUC of 0.851, an ACC of 0.757, and an F1-score of 0.766. Among the baselines, Cosine Similarity achieved the strongest performance, with an AUC of 0.824, an ACC of 0.755, and an F1-score of 0.753. Although the difference in ACC between the two methods was marginal, Pairwise-Semantic Fusion achieved numerically higher AUC and F1-score with relatively lower variability across runs. In particular, the strong performance of Cosine Similarity on the real-world dataset suggests that pairwise semantic relationships between the title and thumbnail and the actual video content can provide useful information for clickbait detection. Pairwise-Semantic Fusion further learns representations of these relationships and integrates them through attention, achieving numerically higher AUC and F1-score than Cosine Similarity. Overall, these results demonstrate the potential of explicitly leveraging the semantic relationships between expectation-forming information and actual video content for clickbait detection on real-world YouTube data. Furthermore, no real-world samples were used to train the proposed model; the real-world data were used only to determine the decision threshold and evaluate the final performance. Nevertheless, Pairwise-Semantic Fusion, trained solely on the synthetic dataset, maintained competitive performance on real-world YouTube videos. This suggests that the semantic relationships learned from the synthetic dataset can, to some

extent, transfer to clickbait occurring in real-world environments, demonstrating the potential of synthetic training data designed around semantic inconsistency for generalization to real-world clickbait detection.

4.3 Analysis

The following analyses were conducted on the synthetic dataset to further investigate the behavior and characteristics of the proposed model. Specifically, we analyze the effects of different branch configurations, the branch-level attention contribution, and the impact of different pooling strategies.

4.3.1 Ablation Study

Table 3. Performance comparison of different pairwise branch configurations

Modality Pair	AUC	ACC	F1
TS	0.829 ± 0.011	0.750 ± 0.024	0.758 ± 0.014
ThS	0.843 ± 0.028	0.737 ± 0.041	0.764 ± 0.021
TK	0.785 ± 0.015	0.647 ± 0.026	0.720 ± 0.015
ThK	0.766 ± 0.028	0.647 ± 0.027	0.704 ± 0.018
TS+ThS	**0.864 ± 0.010**	0.743 ± 0.056	0.771 ± 0.024
TS+TK	0.829 ± 0.010	0.739 ± 0.026	0.751 ± 0.016
TS+ThK	0.835 ± 0.012	0.740 ± 0.042	0.743 ± 0.024
TK+ThS	0.859 ± 0.009	**0.766 ± 0.036**	**0.788 ± 0.022**
TK+ThK	0.793 ± 0.019	0.665 ± 0.021	0.725 ± 0.012
ThS+ThK	0.847 ± 0.008	0.754 ± 0.019	0.770 ± 0.012
ThS+TK+ThK	0.845 ± 0.013	0.755 ± 0.030	0.773 ± 0.019
TS+ThS+ThK	0.858 ± 0.014	0.732 ± 0.042	0.765 ± 0.017
TS+TK+ThK	0.829 ± 0.016	0.716 ± 0.045	0.747 ± 0.017
TS+TK+ThS	0.856 ± 0.021	0.725 ± 0.048	0.766 ± 0.028
All Branches	0.857 ± 0.018	0.735 ± 0.038	0.766 ± 0.024

To investigate the impact of individual modality-pair relationships on clickbait detection performance, we conducted an ablation study using different branch configurations, as summarized in Table 3. Among the single-branch configurations, TS and ThS generally outperformed TK and ThK. In particular, ThS achieved the highest AUC (0.843) and F1-score (0.764), while TS achieved the highest ACC (0.750). These results suggest that semantic relationships involving STT provide relatively informative cues for modeling the relationship between the title or thumbnail and the actual video content. Combining multiple branches generally improved performance, although the degree of improvement varied depending on which branches were combined. Among the two-branch configurations, TS+ThS achieved the highest AUC of 0.864, while TK+ThS achieved the highest ACC (0.766) and F1-score (0.788) across all configurations. However, increasing the number of branches to three or four did not consistently lead to further performance improvements. This indicates that effectively combining complementary semantic relationships may be more important than simply incorporating a larger number of relationships. The All Branches configuration achieved an AUC of 0.857, an ACC of 0.735, and an F1-score of 0.766, showing competitive performance but underperforming some two- and three-branch configurations. One possible explanation is that the relative contribution of each branch may be influenced by the characteristics of the synthetically generated dataset and the potential loss of visual information during the captioning process. Consequently, some modality-pair relationships may provide less informative or redundant features, limiting the benefits of combining all four branches. When considered together with the branch-level attention analysis in Section 4.3.2, these results suggest that individual semantic relationships may contribute differently and highlight the importance of selectively integrating information from multiple branches.

4.3.2 Branch-Level Attention Contribution

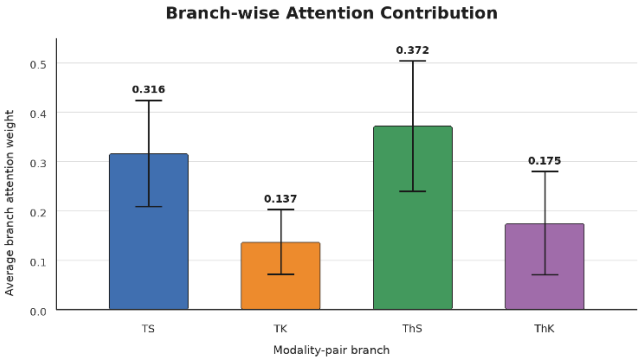


Figure 4. Average branch-level attention weights for each modality pair

We analyzed the branch-level attention weights of four pairwise branches: TK (title–keyframe), TS (title–STT), ThK (thumbnail–keyframe), and ThS (thumbnail–STT). The ThS branch exhibits the highest average attention weight (0.372), followed by TS (0.316), while TK (0.137) and ThK (0.175) show relatively lower contributions. These results indicate that modality pairs involving STT provide stronger signals for detecting semantic inconsistency, while keyframe-based information plays a complementary role.

4.3.3 Comparison of Pooling Methods

Table 4. Performance comparison of different pooling strategies

Pooling	AUC	ACC	F1
Mean	0.824 ± 0.026	**0.744 ± 0.049**	0.755 ± 0.026
Top-k	0.819 ± 0.023	0.704 ± 0.055	0.741 ± 0.024
Top-k(sim)	0.848 ± 0.017	0.736 ± 0.041	**0.770 ± 0.022**
Attention	**0.857 ± 0.018**	0.735 ± 0.038	0.766 ± 0.024

This study compares four pooling strategies to investigate how different sequence summarization methods affect model performance: mean pooling (element-wise averaging) [10], Top-k pooling (magnitude-based selection) [13], Top-k(sim) pooling (similarity-based selection) [14], and attention pooling (query-aware weighting) [9]. For Top-k and Top-k(sim), k was set to 3, as it yielded the best performance among the candidate values tested. Table 4 presents the performance comparison across these methods.

5. Conclusion

This study revisits the problem of YouTube clickbait detection from the perspective of semantic inconsistency. While conventional clickbait detection approaches have primarily relied on sensational expressions or individual features present in titles or thumbnails, this study focuses on the semantic relationships between the expectations formed by titles and thumbnails and the actual content conveyed by videos. To this end, we propose a framework that explicitly models pairwise relationships between expectation-forming modalities and actual content modalities and adaptively integrates these relationships through attention. Furthermore, the model trained solely on synthetic data showed promising generalization to real-world YouTube videos, suggesting that data constructed using the swap-and-generate strategy has the potential to serve as training data for real-world clickbait detection. This strategy provides a potential approach to reducing the burden associated with collecting and manually reviewing large-scale real-world clickbait data and may also be useful for future research in related areas.

Based on these findings, we identify the following directions for future research.

Beyond Textual Unification

In this study, visual information is converted into captions, and a frozen SBERT encoder is used to represent different modalities within a shared textual semantic space. While this approach enables relationships across modalities to be compared within a consistent representation space, visual information that cannot be adequately expressed through language may be lost during the captioning process. For example, facial expressions, the visual presentation of on-screen text, and subtle visual cues in thumbnails may not be fully captured by captions. Future work could incorporate raw visual representations using vision-language models or train the encoder in an end-to-end manner, thereby extending the framework to model semantic relationships that are difficult to capture through textual representations alone.

Modeling More Complex Forms of Semantic Inconsistency and Extending to Long-Form Videos

In this study, clickbait detection is formulated as a binary classification problem between clickbait and non-clickbait. However, even among content classified as clickbait, the degree of inconsistency between the expectations formed by the title or thumbnail and the actual video content may vary substantially. Representing such differences with a single binary label treats relatively mild exaggeration and clickbait that substantially misrepresents the actual content in the same manner. Therefore, future work could move beyond identifying the presence or absence of semantic inconsistency and model its degree on a continuous scale, enabling a more fine-grained analysis of how strongly the presented information exaggerates or misrepresents the actual content. This issue may become even more important when extending the framework to long-form videos. When the overall video content is summarized into a single representation, information may be diluted if content related to the title or thumbnail appears only in specific segments or if semantic inconsistency occurs only during certain parts of the video. Therefore, extending the framework to long-form videos may require not only video-level clickbait prediction but also temporal localization to identify where semantic inconsistency between the expectations formed by the title or thumbnail and the actual content occurs. This extension could enable a more fine-grained form of clickbait detection that captures not only whether content is clickbait, but also the degree of inconsistency and the specific segments in which it occurs.

Generalization to Real-World Environments

In this study, the proposed framework was trained solely on synthetically generated clickbait samples and demonstrated promising performance on real-world YouTube videos. However, the real-world evaluation was conducted on a relatively small-scale dataset, which may not fully represent the diversity of content domains and clickbait strategies encountered in real-world environments. Therefore, future work could evaluate the proposed framework on larger and more diverse real-world

datasets to further validate its generalizability. In addition, although the results demonstrate the potential of synthetic training data, the relative effectiveness of synthetic and real-world training data has not been directly investigated. Future work could conduct additional ablation studies comparing models trained on synthetic data, real-world data, and a combination of both to better understand the contribution of synthetic data to clickbait detection performance.

Conflict of Interest

No potential conflict of interest relevant to this article was reported.

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